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Research paper

A fault diagnosis method for bearings based on multiscale graph convolutional network under non-stationary speed conditions

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ABSTRACT

Current research on intelligent fault diagnosis primarily focuses on constant speed conditions, with less attention given to time-varying speed scenarios. Additionally, relying on single-sensor data limits the robustness and adaptability of intelligent fault diagnosis methods. Therefore, this paper proposes a Multi-Scale Elastic Spatial Convolutional Deep Graph Convolutional Network (ESC-DGCN) Cascaded Fusion Framework for multi-sensor fault diagnosis under time-varying speed conditions. The proposed elastic spatial convolution mechanism adapts to input signals of varying lengths and effectively extracts time-varying speed signal features. The multi-scale graph convolutional network aggregates information from multiple sensor nodes, enhancing feature representation. A cascaded fusion strategy is introduced to efficiently integrate fault features with node features, improving diagnostic performance. Experimental results show that the proposed framework achieves an average diagnostic accuracy of 99.28 % on the Northeast Forestry University dataset and 99 % on two public benchmark datasets, outperforming the best baseline method by 0.72 %, 2.11 %, and 1.3 %, respectively. Experimental results based on datasets from Northeast Forestry University and publicly available datasets validate the effectiveness of the proposed method.

1. Introduction

Rolling bearings are essential components in machinery, widely utilized in critical national strategic sectors such as aerospace, rail transportation, and precision CNC machine tools. The operational condition of rolling bearings directly influences the safety, reliability, and economic performance of mechanical systems [1]. However, bearing failures can lead to equipment damage, system downtime, and pose significant safety risks. Therefore, the efficient and accurate diagnosis of rolling bearing conditions is crucial for ensuring stable equipment operation and reducing maintenance costs [2,3].

Under constant rotational speed conditions, the vibration signals generated by bearing faults typically exhibit periodic impulsive

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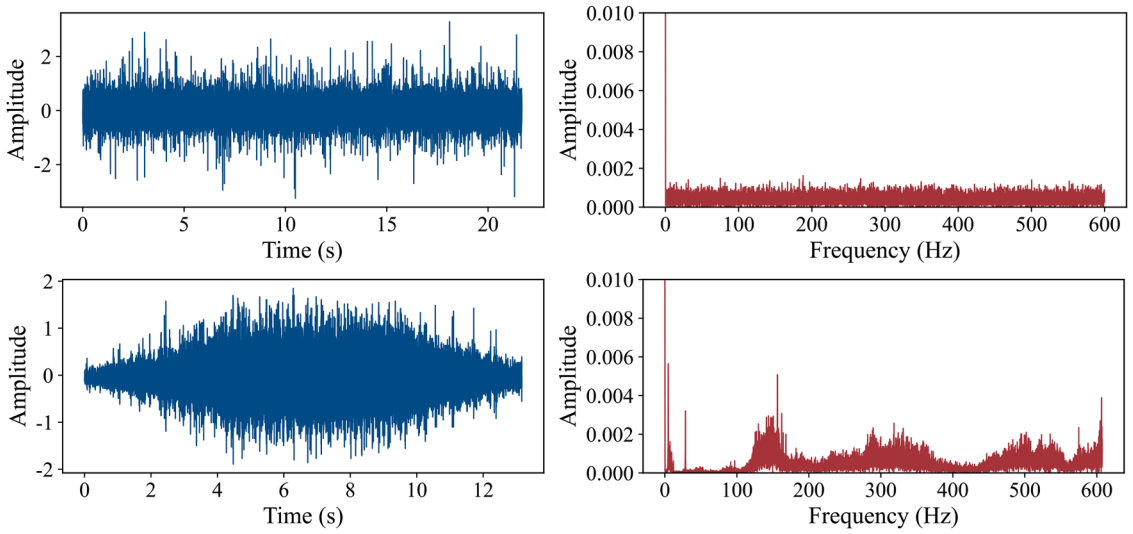


Fig. 1. Time-frequency domain diagrams of stationary signal (upper plot) and time-varying speed signal (lower plot).

characteristics and remain stationary. Traditional diagnostic methods, such as Fourier Transform [4] and envelope analysis [5], effectively extract inherent fault characteristic frequencies. He *et al.* [6] designed a continuous learning framework augmented by feature enhancement and gradient boosting, enabling more adaptive diagnostic capabilities. Tang *et al.* [7] introduced a novel composite diagnosis framework based on a FFT-transformer architecture, employing an attention mechanism to capture features from vibration signals. Ying *et al.* [8] utilized the optimal envelope spectrum (OES) to represent clearer characteristic frequencies, enabling more accurate calculation of the amplitude information of rolling bearings. Zheng *et al.* [9] proposed an improved multi-zone gray wolf optimization algorithm (MZGWO) that uses optimal bandpass filtering to analyze the envelope spectrum for fault information extraction. Wang *et al.* [10] proposed a trackable multi-domain collaborative generative adversarial network (TMCGAN), a generative adversarial network designed with traceability and multi-domain collaboration, to facilitate fault identification in rotating machinery. TMCGAN constructs a multi-domain collaborative adversarial strategy to sequentially learn key feature information from different domains. Existing studies have demonstrated excellent performance under constant-speed conditions but largely rely on the assumption of signal stationarity. In practical applications, equipment such as wind turbines, vehicle power systems, and precision CNC machines often operate with speed fluctuations, resulting in non-stationary vibration signals that weaken the feature extraction and diagnostic capability of traditional methods [11]. Under time-varying speed conditions, issues such as operating disturbances, signal distortion, and class imbalance become even more pronounced, highlighting the urgent need to develop robust intelligent diagnostic methods.

Under non-stationary speed conditions, the fluctuations in speed directly cause the repetitive frequency of bearing fault impacts to no longer be constant but to dynamically change over time [12,13]. This results in vibration signals exhibiting strong non-stationary characteristics, with their frequency spectrum shifting from clear line spectra to a "smearing" phenomenon, where the frequency varies over time, and the energy is dispersed across a wide frequency band, as shown in Fig. 1. Classic spectral analysis methods, such as FFT, which rely on the assumption of signal stationarity, suffer from significant spectral blurring and distortion when processing such signals. This leads to the fault characteristic frequencies being drowned in noise, preventing accurate identification. Most existing methods for diagnosing bearing faults under variable-speed conditions primarily depend on vibration data acquired by a single sensor [14]. Despite extensive research in feature modeling and generalization ability, challenges regarding robustness and limited adaptability remain. To overcome such limitations, researchers have proposed various methods to improve diagnostic performance under complex operating conditions. Gao *et al.* [15] designed a semi-supervised diagnostic model that combines feature perturbation with decision fusion, enabling effective adaptation to complex conditions through a dual-model collaboration mechanism. Kim *et al.* [16] developed a domain generalization framework that embeds prior knowledge into a neural network, making decisions for non-stationary signals based on the order of fault features. Shi *et al.* [17] built a similar framework by embedding prior knowledge into the neural network to discriminate non-stationary signals based on feature fault sequences. Liu *et al.* [18] introduced an imbalance-oriented multi-domain generalization strategy, incorporating a specific loss function to improve the model's sensitivity to fault characteristics in complex scenarios. Pan *et al.* [19] proposed a novel intelligent diagnosis method based on the Semi-Supervised Matrixized Graph Embedding Machine (SMGEM), which first captures the geometric similarity between labeled and unlabeled samples, while employing a weighted nuclear norm to emphasize the significance of singular values. Cheng *et al.* [20] introduced a novel deconvolution method that introduces the concept of generalized envelope into the convolution operation and employs the hybrid Ramanujan Fourier transform to analyze signals. Zhi *et al.* [21] proposed the Local Entropy-Selected Scaling Extracted Chirplet Transform, which adaptively selects multiple chirp frequencies at the same time center, thereby significantly enhancing bearing fault diagnosis performance under time-varying rotational speed conditions. Single-sensor fault diagnosis methods offer advantages in practical applications, such as ease of deployment, fast response, and low computational cost [22,23]. However, as operating

conditions become increasingly complex, relying solely on information obtained from a single sensor provides limited dimensionality, making it difficult to comprehensively and accurately characterize the equipment operating state. This limitation has also been highlighted in recent studies, underscoring the need to explore multi-source information fusion and more robust intelligent diagnosis methods to enhance fault identification performance under complex condition.

To address these limitations, researchers have shifted their focus to multi-sensor data-driven methods. Multi-sensor systems collect vibration signals from different locations and angles, which improves signal stability and provides richer fault information. Especially under non-stationary speed conditions, multi-sensor data fusion helps alleviate the sensitivity of single sensors to operating condition fluctuations, improving diagnostic accuracy and generalization ability. In order to empower the model with multi-source representational capability, Li *et al.* [24] introduced a MixStyle network enhanced by the SE attention mechanism, which dynamically allocates weights via the attention mechanism to capture signal features across channels. Li *et al.* [25] introduced a graph neural architecture capable of integrating both time- and frequency-domain representations through a multi-view perspective, using convolutional neural networks and FFT to convert node features into two distinct feature spaces, forming multi-view aggregation of sensor information. To enhance automatic diagnosis of bearing defects, Mao *et al.* [26] introduced a fusion model based on residual convolution and multi-sensor data, effectively capturing global features from sound and vibration sources. Qin *et al.* [27] introduced an improved multi-agent deep reinforcement learning collaborative diagnosis network to enhance classification capability by customizing the interaction environment and assigning each agent a diagnostic subtask for specific components. Zhou *et al.* [28] developed a novel multi-branch horizontal enhancement network to capture local degradation features and dependencies at different scales. Tao *et al.* [29] developed a convolutional neural network that processes multi-sensor signals in parallel, capturing fault features under different sensor information. However, current fusion methods (such as data concatenation or weighted averaging) often fail to fully consider the spatiotemporal correlations between different sensors, making it difficult to capture and model the complex interactions between sensor channels [30]. Moreover, traditional convolution methods typically use fixed-size and fixed-shape kernels, performing sliding-window feature extraction on the input data. The receptive field of such methods is rigid, meaning it can only learn predefined, local patterns. In time-varying speed conditions, the shape and scale of bearing fault signals are distorted and stretched as speed and load change. A fixed convolution kernel has difficulty adapting to these geometric deformations, leading to significant performance degradation when facing varying operating conditions. Essentially, this rigid structure forces the model to measure fault features with a single "ruler", limiting its flexibility and generalization ability.

In response to the aforementioned limitations, this paper proposes a multiscale elastic graph convolutional network framework for fault diagnosis based on cascaded fusion. The proposed method comprises three key components: autoregressive convolutional feature extraction, multi-receptive field graph convolutional network, and a cascaded fusion strategy. To address the "smearing phenomenon" in time-varying speed signals, the first layer employs an autoregressive convolutional module. This design enhances the model's ability to capture temporal dependencies in fault signals. Next, a multi-receptive field design is applied to capture fault patterns at different scales, thereby constructing feature interrelationships. Finally, the cascaded fusion strategy integrates the local feature extraction capability of convolutional neural networks (CNNs) with the global relationship modeling power of graph neural networks (GNNs). This complementary representation enhances the model overall feature learning. Experiments on a laboratory dataset and two public datasets validate the superiority of the proposed method. The main contributions of this paper are summarized as follows:

(1) Introduce autoregressive modeling to dynamically adjust the receptive field.
(2) Construct dynamic topological relationships to capture multi-scale faults.
(3) Fuse autoregressive and graph convolution for hierarchical feature learning.

The remainder of this paper is organized as follows. Section 2 presents the theoretical background and related work. Section 3 provides a detailed description of the proposed framework and the implementation of each component. Section 4 presents experimental validation using datasets from the Northeast Forestry University laboratory, Tsinghua University, and Shandong University of Science and Technology. Finally, Section 5 concludes the paper and discusses future work directions.

2. Theoretical background

2.1. Autoregressive modeling

Autoregressive (AR) modeling [31] is a classical technique in time series analysis. Its core idea is that the current value of a signal can be represented as a linear combination of its past values, implying that the current state depends on its historical states. This modeling approach is capable of accurately capturing the inherent temporal structure and dynamic evolution patterns of data, and finds extensive use in domains including speech analysis, economic prediction, and mechanical fault identification.

An autoregressive model of order one (AR(1)) can be expressed as:

$$x_t = \phi_1 x_{t-1} + \varepsilon_t \tag{1}$$

Where, x_t denotes the observed signal value at time t, ϕ_1 is the regression coefficient, and ε_t represents the white noise disturbance term. This can be generalized to an AR model of order p as:

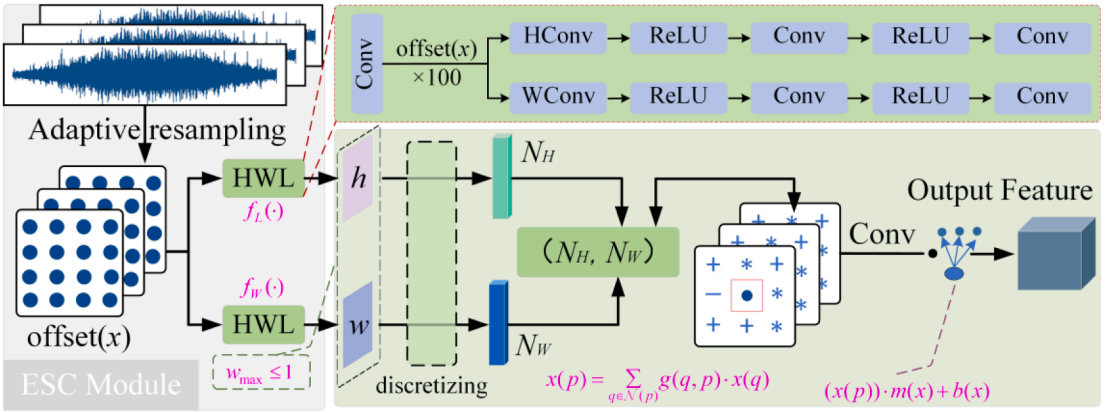


Fig. 2. Elastic spatial convolution module.

$$x_t = \sum_{i=1}^{p} \phi_i x_{t-i} + \varepsilon_t \tag{2}$$

Where, ϕ_i represents the regression coefficient of the i th lag term, and it is assumed to follow an independent and identically distributed process with zero mean and a variance ofσ^2.

The AR model has a clear analytical expression in the frequency domain, and its power spectral density is given by:

$$P(\omega) = \frac{\sigma^2}{\left|1 - \sum_{i=1}^{p} \phi_i e^{-j\omega i}\right|^2} \tag{3}$$

The AR model, as an all-pole filter, can effectively capture the spectral peaks of a signal through the positions of its poles, allowing for precise estimation of its dominant frequency components. This characteristic makes it particularly suitable for modeling vibration signals with distinct periodicity, such as the periodic impact responses generated by bearing faults.

Although the traditional AR model demonstrates excellent time-series modeling capabilities, its linear assumptions limit its adaptability to complex fault features, such Nas non-stationary, nonlinear, and abrupt patterns. Building upon traditional autoregressive modeling, this study integrates the strengths of CNNs by introducing the concept of "autoregressive convolution" to simulate the AR process's capability of modeling local temporal dependencies. Specifically, the AR model captures autocorrelation features in time series, while the CNN effectively handles spatial relationships in high-dimensional data. By combining the advantages of these two models, this study captures richer local dependency information in temporal signals, thereby improving fault diagnosis accuracy. The incorporation of the AR process enables the network to better address temporal dependencies in signals, while the CNN extracts local features, ensuring their complementarity and thus providing a more effective solution for complex fault diagnosis tasks under variable speed conditions.

2.2. *Graph convolutional network (GCN)*

Graph Convolutional Networks (GCNs) are a class of deep learning models specifically designed for graph-structured data, and they have been widely applied in various fields such as social network analysis, bioinformatics, traffic flow prediction, and industrial fault diagnosis [32,33]. In a standard GCN framework, a graph is represented as $G = (V,E)$, where V denotes the node set and E indicates the edge set. The relationships between nodes are encoded by an adjacency matrix A.

The fundamental idea of GCNs is to perform convolutional aggregation of node features based on the graph's adjacency structure—updating the representation of a central node by propagating information from its neighboring nodes. A classical GCN layer can be formulated as:

$$H^{(l+1)} = \sigma\left(\widehat{D}^{-1/2}\widehat{A}\widehat{D}^{-1/2}H^{(l)}W^{(l)}\right) \tag{4}$$

Where,$\widehat{A} = A + I$ is the adjacency matrix with self-loops added, $\widehat{D}$ is the corresponding degree matrix, $H^{(l)}$ is the node feature representation at the l-th layer, $W^{(l)}$ is the learnable weight matrix, and $\sigma(\cdot)$ is the activation function.

The standard GCN model is limited to first-order neighbor information propagation, making it difficult to capture broader structural relationships. Therefore, researchers have proposed spectral graph convolution based on Chebyshev polynomial approximation (ChebNet), which introduces higher-order neighborhoods to achieve a larger receptive field. The mathematical formulation is expressed as follows:

$$x' = \sum_{k=0}^{K} \theta_k T_k(\widetilde{L}) x \tag{5}$$

Where,$\widetilde{L}$ is the normalized Laplacian matrix, $T_k(\cdot)$ represents the Chebyshev polynomial, and θ is the learnable parameter.

3. Method

3.1. Elastic spatial convolution (ESC)

Traditional convolution has a fixed and uniform receptive field, which limits its ability to model non-rigid deformations or complex structural targets. Inspired by variable convolutions [34], this paper proposes the Elastic Spatial Convolution (ESC) module. Unlike point-to-point offset learning, ESC adopts a scale-guided mechanism that dynamically adjusts the receptive field size and sampling pattern for each position in the spatial dimension. This adaptive adjustment of the sampling range enables more effective modeling of spatially deformed targets. The specific structure is shown in Fig. 2.

First, the one-dimensional vibration signal is adaptively resampled and mapped into a two-dimensional representation, fully preserving its sequential, frequency-domain, and statistical properties. Let the input features be:

$$x \in \mathbb{R}^{B\times C\times H\times W} \tag{6}$$

Where, B is the batch size, C is the number of channels, H s the height of the feature map, and W is the width of the feature map.

The ESC module first predicts the scale factors of the receptive field in the vertical (height) and horizontal (width) directions through two convolutional sub-networks. The mathematical formulation is as follows:

$$\begin{cases} H = \sigma(f_L(x)) \cdot (h_{\max} - 1) + 1 \\ W = \sigma(f_W(x)) \cdot (w_{\max} - 1) + 1 \end{cases} \tag{7}$$

Where,$\sigma(\cdot)$ denotes the Sigmoid activation function, $h_{\max}$,$w_{\max}$ is the predefined maximum receptive field size, and$f_L(\cdot)$, $f_W(\cdot)$ are convolutional networks used to predict the scale factors in the height and width directions, respectively.

Next, the predicted scale factors are spatially averaged and discretized by rounding down to obtain integer values for the receptive field size:

$$\begin{cases} N_H = 2 \cdot \overline{H} + 1 \\ N_W = 2 \cdot \overline{W} + 1 \end{cases} \tag{8}$$

Where,N_H, N_Wrepresent the discretized receptive field height and width. $\overline{H} = 1/H \times W \cdot \sum_{i,j} H_{i,j}$,$H_{i,j}$denotes the predicted vertical scale factor at spatial position (i, j), and $W_{i,j}$ denotes the predicted horizontal scale factor at the same position. The $\lfloor \cdot \rfloor$ operator denotes the floor operation to ensure the receptive field sizes are integers.

The ESC module dynamically selects the convolution kernel of the appropriate size for feature extraction based on (N_H, N_W). The ESC module applies elastic sampling to the standard regular sampling points $P_n \in \mathbb{R}^{2N}$ through a scale transformation. The coordinates of the sampling points are defined as:

$$p = p_0 + offset(x) \cdot p_n \tag{9}$$

Where,$p_0$represents the center coordinates of the convolution kernel, and offset(x) is the scale factor learned from the input features. Unlike Deformable Convolution, which performs point-wise free offsets, the offsets in ESC are based on global size adjustments, ensuring the regularity and stability of the sampling structure.

Since the sampling points are typically floating-point coordinates, ESC uses bilinear interpolation to sample from the input features:

$$x(p) = \sum_{q \in \mathcal{N}(p)} g(q,p) \cdot x(q) \tag{10}$$

Where,$\mathcal{N}(p)$ denotes the four integer grid neighbors surrounding the sampling point p, and$g(q,p)$represents the interpolation weights.

After the sampled features are processed by the corresponding convolution kernels, ESC introduces a modulation mechanism to enhance the response to key feature regions. The expression is defined as:

$$y = \mathrm{Conv}_{N_H \times N_W}(x(p)) \cdot m(x) + b(x) \tag{11}$$

Where,$m(x)$and$b(x)$are the modulation factor and bias term generated from the input features, respectively. By dynamically adjusting the output amplitude, they enhance the model's flexibility in feature representation.

3.2. Deep graph construction factorization (DGCF)

In practical industrial applications, the structural relationships between nodes are often implicitly present, making them difficult to explicitly capture. Especially in multi-sensor time-series signal modeling tasks, fixed graph structures struggle to accurately

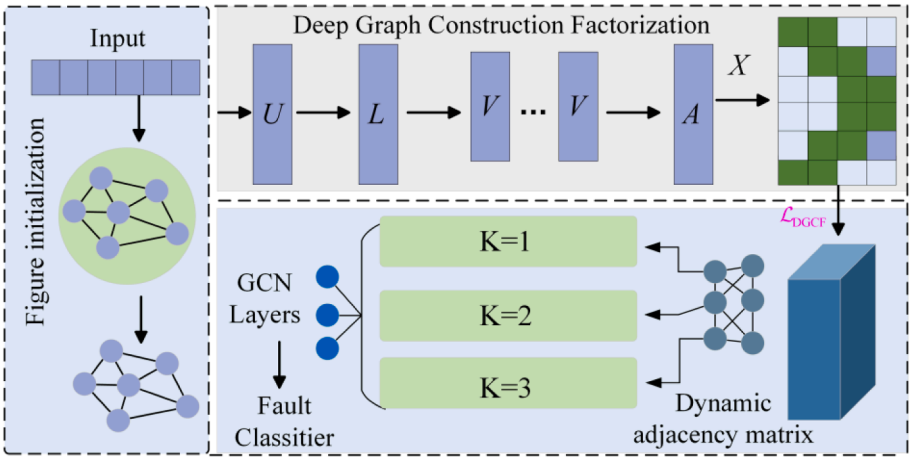


Fig. 3. Deep Graph Construction Factorization.

characterize the latent dependencies between nodes. Therefore, this paper introduces a learnable non-negative factor matrix to dynamically construct the graph structure and designs the DGCF module, achieving joint optimization of the graph structure and node representations, the specific structure is illustrated in Fig. 3.

The module performs layer-wise factorization on the original input feature matrix $X \in \mathbb{R}^{N\times F}$and the graph structure matrix $A \in \mathbb{R}^{N\times N}$, learning the underlying low-rank non-negative factors:

$$A \approx U_1 U_2 \cdots U_L V, \quad X \approx U_1' U_2' \cdots U_L' V' \tag{12}$$

Where, $U_1, U_2, \ldots, U_L \in \mathbb{R}^{r_i \times r_{i+1}}$ denotes the layer-wise non-negative factor matrix, with each layer compressing the high-dimensional structure into a lower-dimensional one. $V \in \mathbb{R}^{r_{L+1}\times N}$ represents the factor matrix at the bottom layer, used to reconstruct the adjacency matrix A. $U_1', U_2', \ldots, U_L' \in \mathbb{R}^*$ refers to the multi-layer non-negative factors in the feature space.

The above factorization not only provides a compact representation of node attributes but also models the underlying topological structure. Specifically, the factor matrix U_i, together with the node set V, determines the final graph connectivity. The adjacency matrix is constructed as follows:

$$\widehat{A} = \mathrm{ReLU}\left(U_1 \cdots U_L V V^{\mathrm{T}} U_L^{\mathrm{T}} \cdots U_1^{\mathrm{T}}\right) \tag{13}$$

Where, $\widehat{A}$ denotes the reconstructed adjacency matrix obtained through factorization and nonlinear mapping, ReLU represents the activation function, $VV^{\top}$ denotes the low-rank matrix, captures the latent relationships among nodes. This approach can be regarded as an end-to-end trainable graph generator, which dynamically adjusts the topology via backpropagation of the training objective, thereby endowing the GNN with flexible structure-aware capability.

To ensure the sparsity of the graph structure, we impose non-negative constraints and sparse regularization on the above matrices:

$$\mathscr{L}_{\mathrm{DGCF}} = \| A - \widehat{A} \|_F^2 + \| X - \widehat{X} \|_F^2 + \alpha \cdot \left(\sum_i \|U_i\| + \| V \|\right) \tag{14}$$

Where, $\widehat{X} = U_1' \cdots V'$ denotes the feature representation reconstructed through the factorization.

After obtaining the dynamic adjacency matrix, we further use multi-order Chebyshev graph convolution to extract structural embedding features:

$$\begin{cases} Z = \overline{\sum_{k=1}^{K}} Z^{(k)} \\ Z^{(k)} = \mathrm{ChebConv}_k(X, \widehat{A}) \end{cases} \tag{15}$$

Where, $\mathrm{ChebConv}_k(\cdot)$ denotes the k-th order Chebyshev graph convolution operation, used to extract local features on the graph structure. $Z^{(k)} \in \mathbb{R}^{N\times d}$ denotes the embedding vector of a node produced by the k-order graph convolution operation, with the embedding dimension of each node being d.

By stacking multiple scales (different orders) of graph convolution operations, the model can simultaneously capture both local and global graph structural information, enhancing its ability to model complex patterns.

Finally, to prevent model overfitting, a loss function is designed for regularization, and the mathematical formulation is as follows:

$$L = \lambda_1 \cdot \mathscr{L}_{\mathrm{DGCF}} + \lambda_2 \cdot \mathscr{L}1 \tag{16}$$

Where, λ_1, λ_2 denotes the weight associated with each term of the loss function.

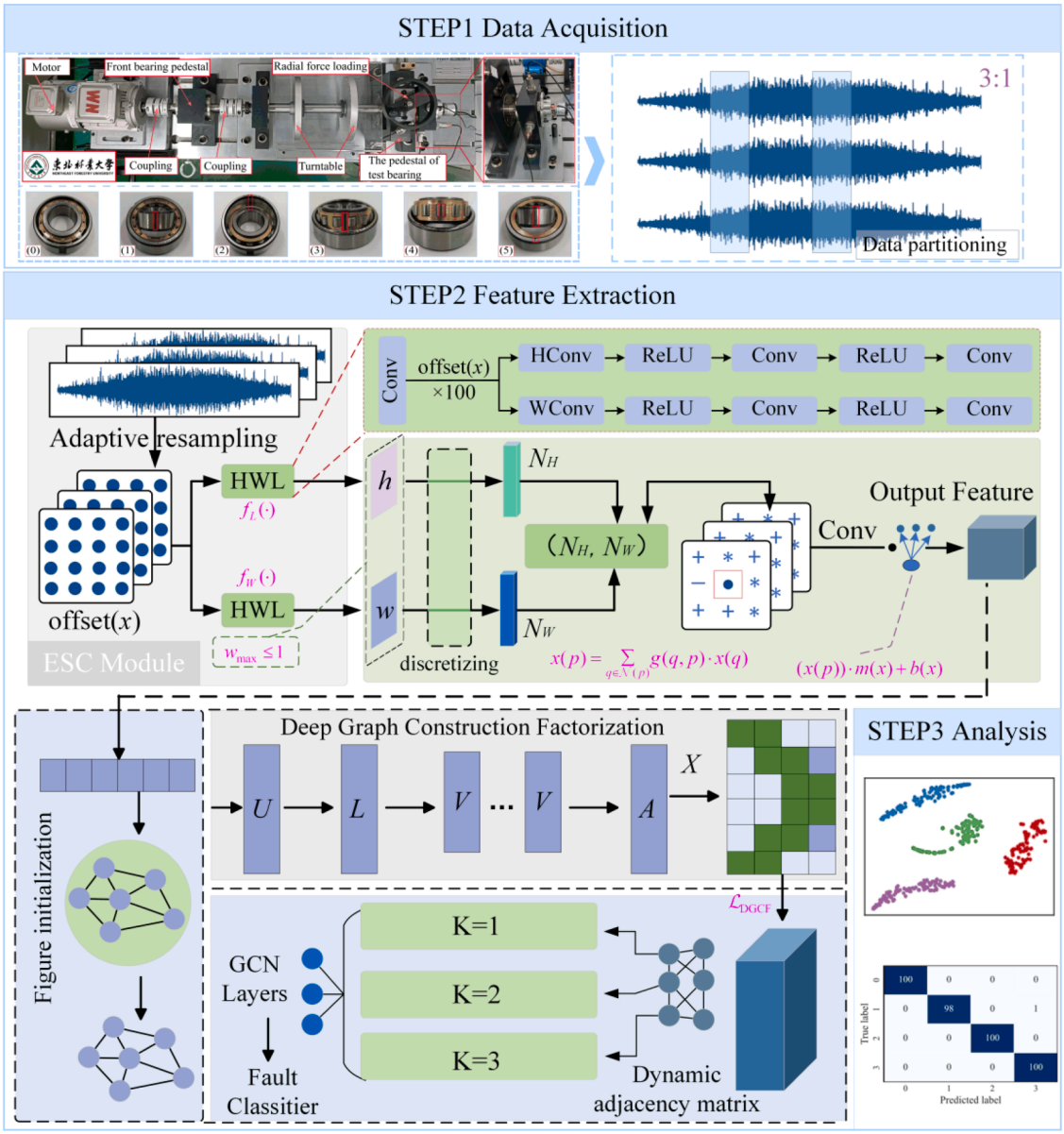


Fig. 4. ESC-DGCN fault diagnosis framework.

3.3. *Proposed method*

To address the challenges of multi-sensor data-driven bearing fault diagnosis under time-varying speed conditions, we design a multi-scale Elastic Graph Convolutional Network framework and propose the ESC-DGCN model, which demonstrates outstanding fault identification ability. The practical use of the proposed framework in bearing fault scenarios is introduced in this section. Fig. 4 presents a detailed workflow of the ESC-DGCN-based intelligent diagnostic method.

The detailed steps are outlined below:

Step 1: Data Collection: Acquire multi-sensor time-varying speed vibration data from the test bench equipment, ensuring that the data covers different fault states.

Step 2: Data Preprocessing: Perform data cleaning, denoising, and normalization on the collected data to ensure data quality. The dataset is allocated to training and test sets based on a 3:1 ratio.

Step 3: Feature Extraction: The ESC module performs elastic receptive field convolution through scale transformation, using bilinear interpolation and modulation mechanisms to enhance the response of key regions. The DGCF module constructs a dynamic graph structure from the extracted output features and performs multi-order graph convolution.

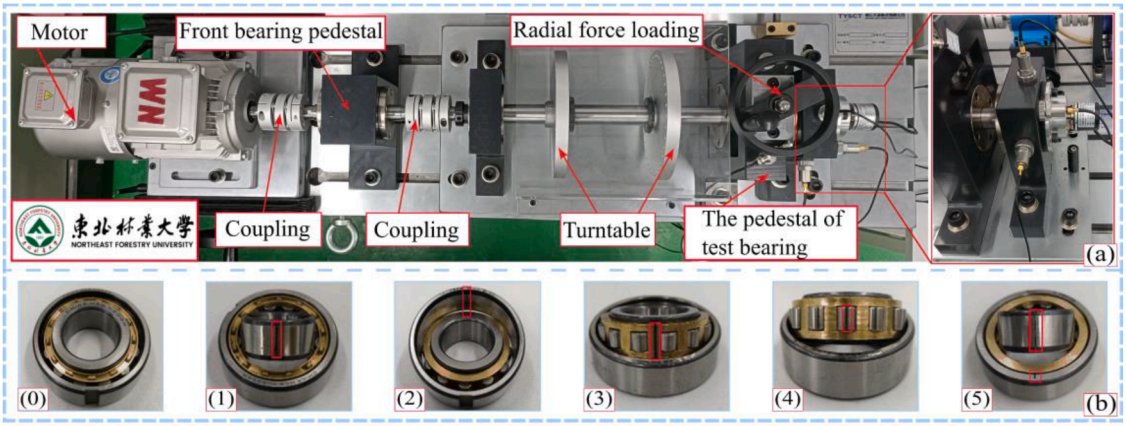


Fig. 5. Detailed structure of the NEFU test rig.

Table 1
NEFU Bearing Dataset Overview.

Health Status	Label	Health Status	Label
Normal Bearing	0	Cage Fault	3
Inner Race Fault	1	Rolling Element Fault	4
Outer Race Fault	2	Inner & Outer Race Fault	5

Step 4: Bearing Fault Diagnosis: The ESC-DGCN model is trained using the training set, optimizing the model parameters during the training process, and evaluating the model performance on the test set.

4. Experimental validation

This section presents experimental studies from multiple perspectives to evaluate the effectiveness of the proposed ESC-DGCN in bearing fault identification. All experiments share the same configuration: a batch size of 64, use of the Adam optimizer with adaptive learning rate adjustment starting from 0.0001, and a training duration of 200 epochs. To ensure a fair comparison, all competing approaches adopt the same hyperparameter settings as the proposed method.

4.1. Comparison methods

To validate performance, the proposed technique is benchmarked against various advanced methods for fault detection, with the specific details outlined as follows.

(1) CNN Model for capturing spatial and temporal features in signals [35]: Utilizes fixed convolutional kernels to capture useful features from raw vibration signals, making fault diagnosis more efficient.
(2) AT-ICNN Model with Real and Imaginary Part Convolutional Kernels [36]: Independently processes key time-frequency information from each channel, enhancing the richness of feature representations.
(3) PeriodicMFD Model for Multi-source Cross Speed Data [37]: Captures periodic features by constructing periodic samples, addressing the inconsistency in feature dimensions across multi-source samples.
(4) CNN-LSTM Model Combining Convolutional Neural Networks and Long Short-Term Memory Networks [38]: CNN extracts spatial features from the data, while LSTM uncovers hidden temporal patterns.
(5) DAGCN Model Using Graph Convolutional Networks and Maximum Average Difference [39]: Extracts structural feature relationships from samples, constructing instance graphs for modeling via graph convolutional networks.

4.2. Northeast forestry university time-varying speed bearing dataset

The test platform developed at Northeast Forestry University (NEFU) is designed to emulate bearing behavior under variable-speed conditions across different health states. It comprises a 0.75 kW AC variable-frequency motor, dual rotor imbalance disks, a radial load module, a bearing fixture for fault simulation, a speed encoder, and three ICP accelerometers installed at distinct locations (the setup is illustrated in Fig. 5(a)). During data collection, the sampling frequency is set to 30 kHz, and the motor speed gradually varies from 0 to 900 r/min, creating test samples with speed fluctuation characteristics to evaluate the fault diagnosis capability of the algorithm under time-varying speed conditions.

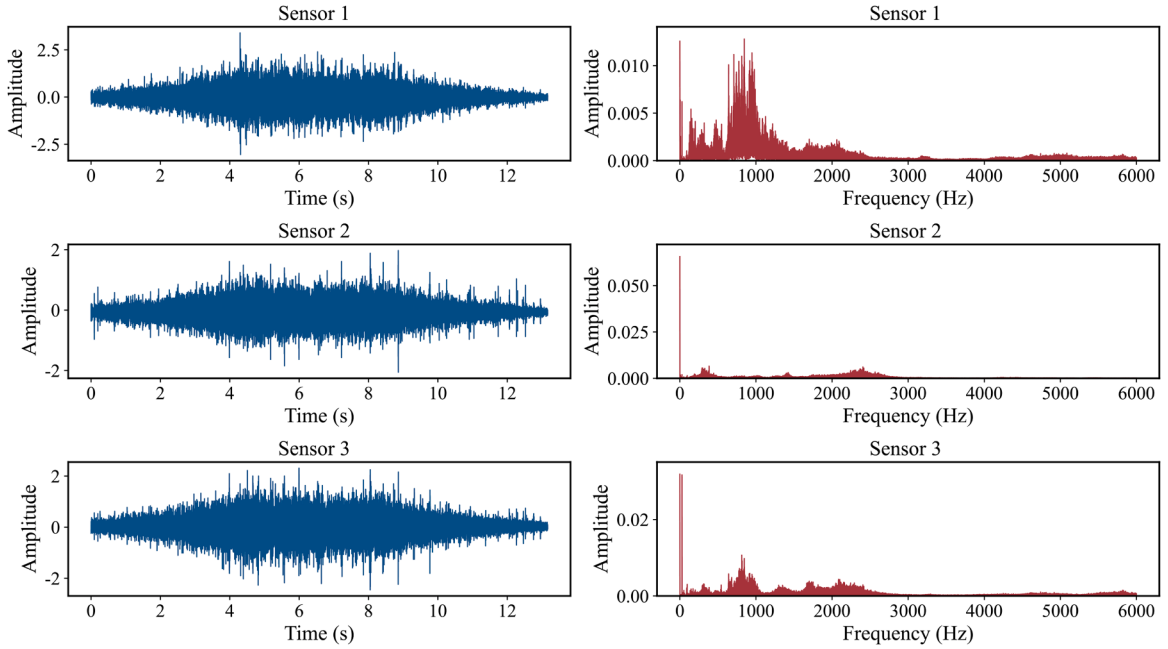
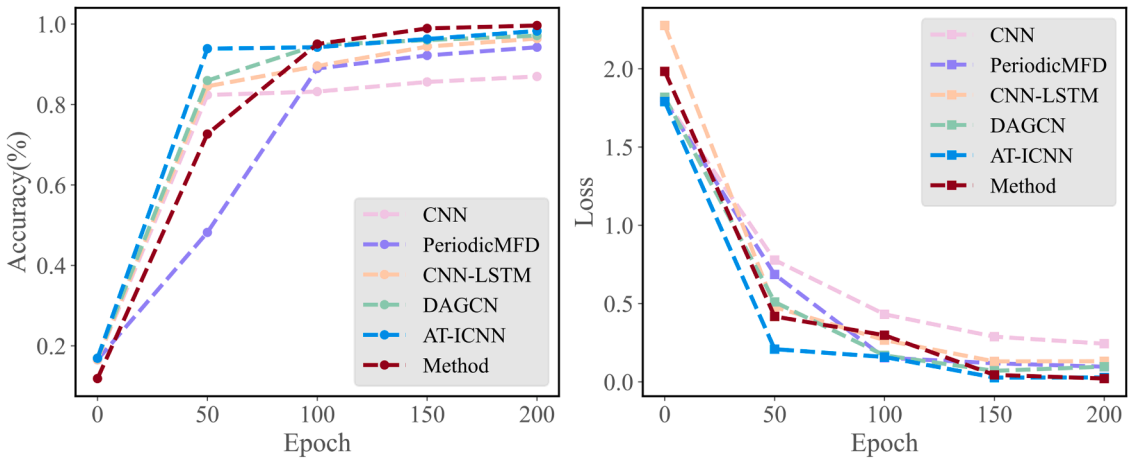


Fig. 6. Visualization of NEFU vibration signal and its frequency-domain curve.



Method	CNN	PeriodicMFD	CNN-LSTM	DAGCN	AT-ICNN	Method
Accuracy	86.61±0.01	95.94±0.02	96.36±0.01	97.48±0.003	98.56±0.01	**99.28±0.002**

Fig. 7. Comparison of training accuracy and loss among different models.

The bearing dataset with different health statuses is shown in Fig. 5(b). It consists of 5 fault states and 1 normal state, totaling 6 categories. The bearing health statuses and corresponding labels are presented in Table 1.

To visually demonstrate the vibration signal features under time-varying speed conditions, the vibration signal of the bearing with label 2 is selected for visualization, as shown in Fig. 6. Each sample consists of 1024 consecutive data points obtained through overlapping window sampling. For every bearing condition, 150 samples are generated, of which 70 % are allocated for training and the remaining 30 % for testing. Each model is tested with 10 repeated trials to avoid randomness, and the performance metrics are collected after 30 epochs of training (all subsequent experiments follow the same settings).

The average accuracy and standard deviation across models are presented in Fig. 7. The six models (CNN, PeriodicMFD, CNN-LSTM, DAGCN, AT-ICNN, and the proposed Method) were each run 10 times, and their average training accuracies were 86.61 %, 95.94 %, 96.36 %, 97.48 %, 98.56 %, and 99.28 %, respectively. Throughout training, model accuracy improves steadily and loss decreases until both metrics converge in the last 30 epochs. Throughout the entire training process, the proposed approach achieves

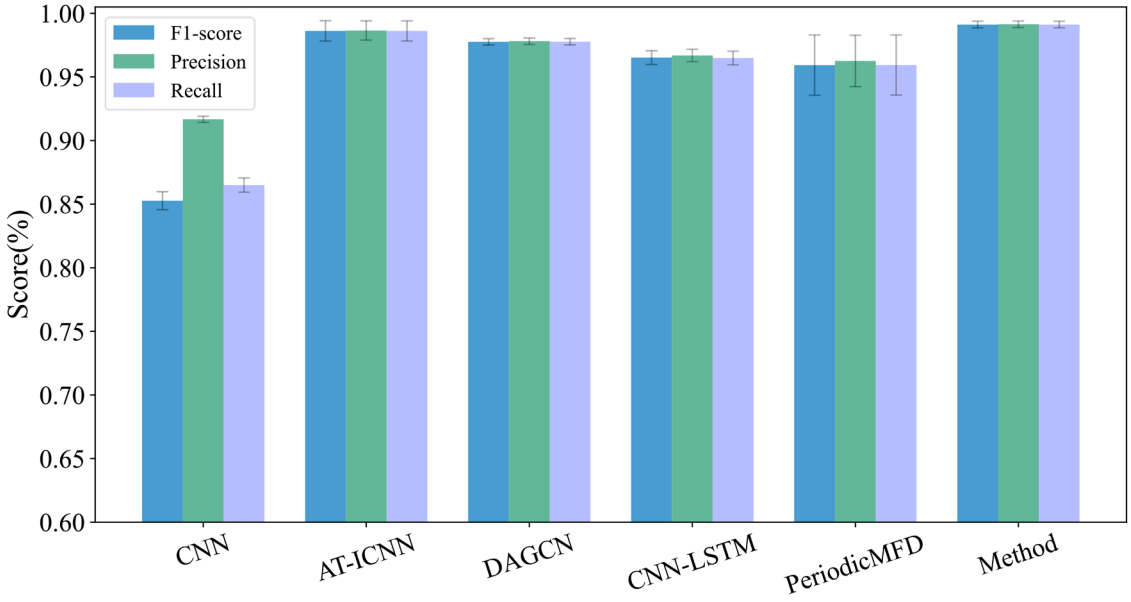


Fig. 8. Comparison of evaluation metrics for different models.

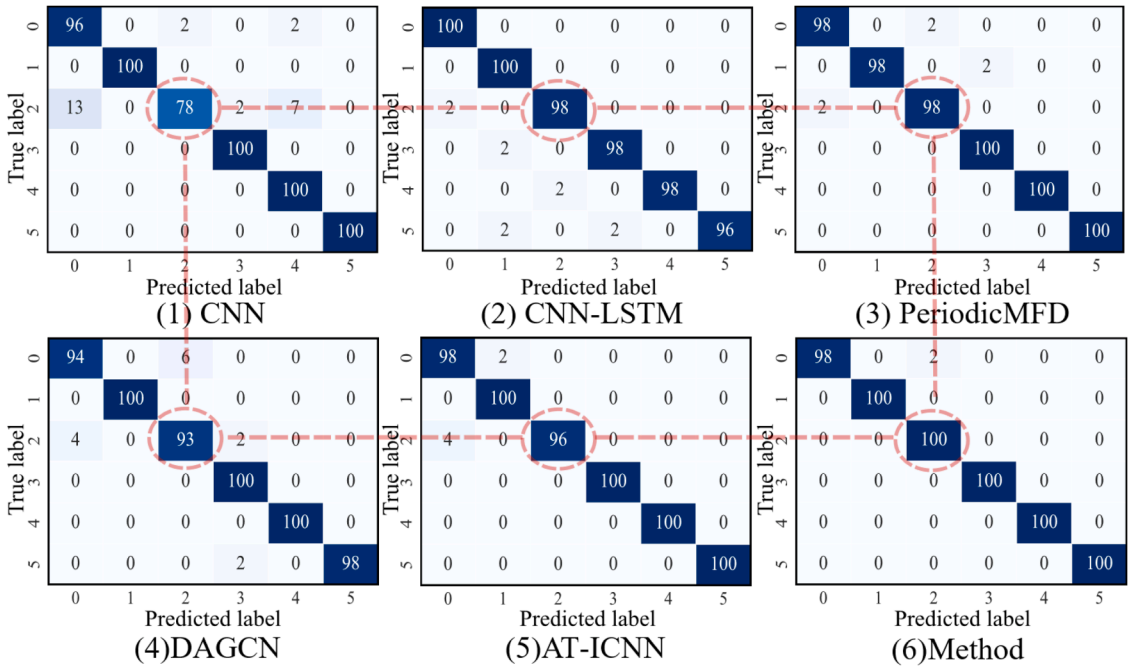


Fig. 9. Models confusion matrix analysis.

superior performance, with quicker loss decline and smoother convergence. The performance of the proposed approach is further assessed through a comparison of F1-score, Precision, and Recall across different models, as illustrated in Fig. 8.

As shown in Fig. 8, the proposed method outperforms the comparison models across all evaluation metrics, indicating that it not only achieves higher classification accuracy but also exhibits excellent recall capability and generalization performance. This allows for a more comprehensive and accurate identification of bearing fault types. These results further validate the effectiveness of the proposed model architecture and training strategy.

As a key evaluation method, the confusion matrix offers detailed insight into both accurate and misclassified results among various categories. As shown in Fig. 9, Each confusion matrix is normalized to an accuracy range of 0–100 %. In the matrix, rows represent the predicted labels and columns represent the true labels. The diagonal elements indicate the classification accuracy for each class (i.e.,

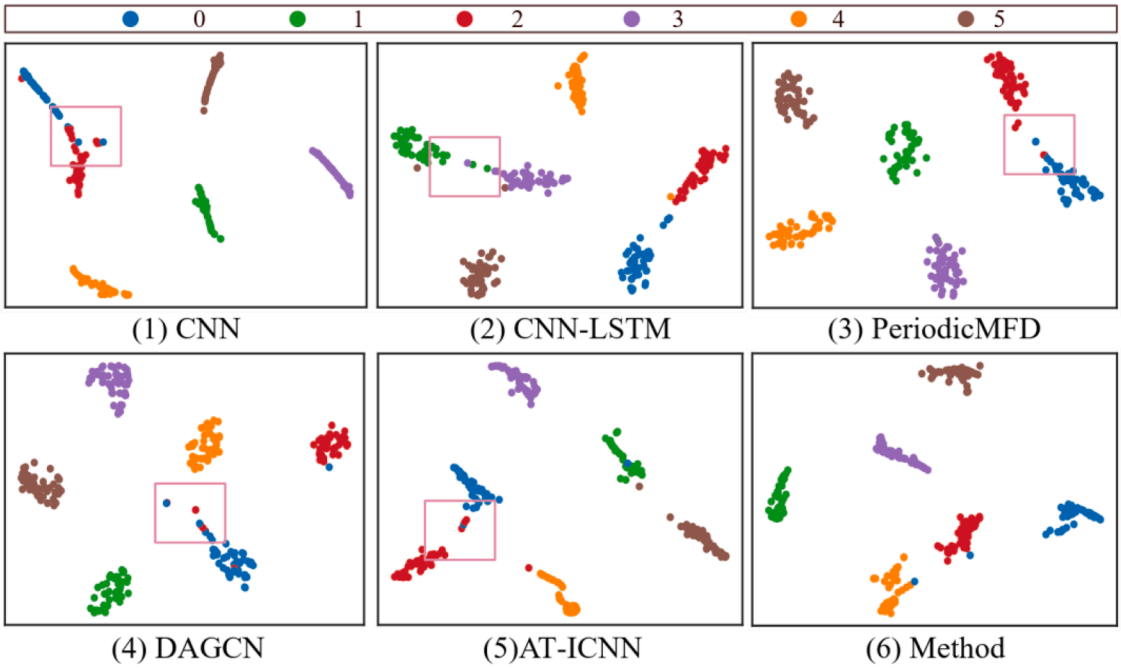


Fig. 10. t-SNE visualization of different models.

Table 2
Ablation experiments on NEFU.

Method	Accuracy(%)	F1-score(%)	Recall(%)
Baseline-CNN	86.61	86.58	86.72
ESC-only	94.23	93.90	94.05
DGCF-only	95.32	94.85	95.10
ESC-DGCN w/o Multi-Scale GCN	97.20	97.69	97.11
ESC-DGCN	**99.28**	**99.15**	**99.12**

the percentage of correctly classified samples), while the off-diagonal elements represent the percentage of misclassified samples. As shown in Fig. 9, all the compared models exhibit misclassification for label 2 (indicated by the dashed lines), indicating that recognizing this fault type remains challenging under non-stationary speed conditions. In contrast, the proposed ESC-DGCN demonstrates excellent performance across all labels, highlighting its robustness and accuracy under complex operating conditions. Diagonal values show the percentage of correctly classifications per class, while off-diagonal values reflect the percentage of misclassification rates. The proposed method delivers excellent classification performance, as evidenced by the experimental results. To assess the model's capability in feature learning, t-SNE was applied to visualize the output representations in a lower-dimensional space, as illustrated in Fig. 10. Even under complex time-varying speed conditions, the method demonstrates the ability to extract discriminative features and accurately identify fault types.

To systematically evaluate the role of each module in the model, ablation experiments were conducted to quantitatively analyze the contributions of the ESC module, the DGCF module, and key sub-mechanisms. Using the same experimental parameters as above, the model was trained for 200 epochs and repeated ten times to obtain the average results, as shown in Table 2.

The Baseline-CNN model achieved an accuracy of only 86.61 %, indicating that conventional convolution alone is insufficient to fully capture the feature distributions under time-varying speed conditions. After introducing the ESC module, the accuracy increased to 94.23 %, demonstrating that elastic spatial convolution can effectively expand the receptive field and enhance responses in key regions. The DGCF-only module further improved the accuracy to 95.32 %, validating the importance of dynamic graph structure modeling in characterizing the latent relationships among multiple sensors. When the two modules were combined without incorporating multi-scale graph convolution, the model still achieved 97.20 %, suggesting a complementary effect between ESC and DGCF. Finally, the complete ESC-DGCN framework attained an accuracy of 99.28 % along with high F1-score and Recall on this dataset, significantly outperforming all ablated variants and fully confirming the effectiveness of the proposed method on the NEFU dataset.

4.3. Tsinghua university variable operating condition bearing dataset

To assess how the proposed approach behaves under various time-varying speed scenarios, the Tsinghua University (THU) variable

Table 3
Overview of the THU bearing dataset.

Health Status	Label	Health Status	Label
Healthy Gear + Healthy Bearing	0	Gear Tooth Break + Outer Race L	4
Gear Tooth Break + Inner Race L	1	Gear Tooth Break + Outer Race M	5
Gear Tooth Break + Inner Race M	2	Gear Tooth Break + Outer Race H	6
Gear Tooth Break + Inner Race H	3		

Table 4
Detailed fault parameters of the THU bearing.

Bearing Fault Parameters	Fault Severity		
	Light (L)	Medium (M)	Heavy (H)
Inner Race Fault Width	0.1mm	0.3mm	0.5mm
Outer Race Fault Width	0.1mm	0.3mm	0.5mm

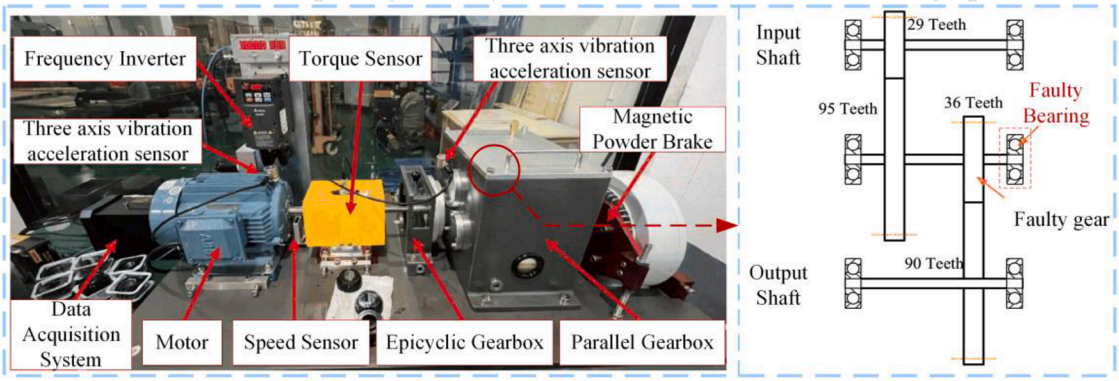


Fig. 11. Detailed structure of the THU fault diagnosis test rig [40].

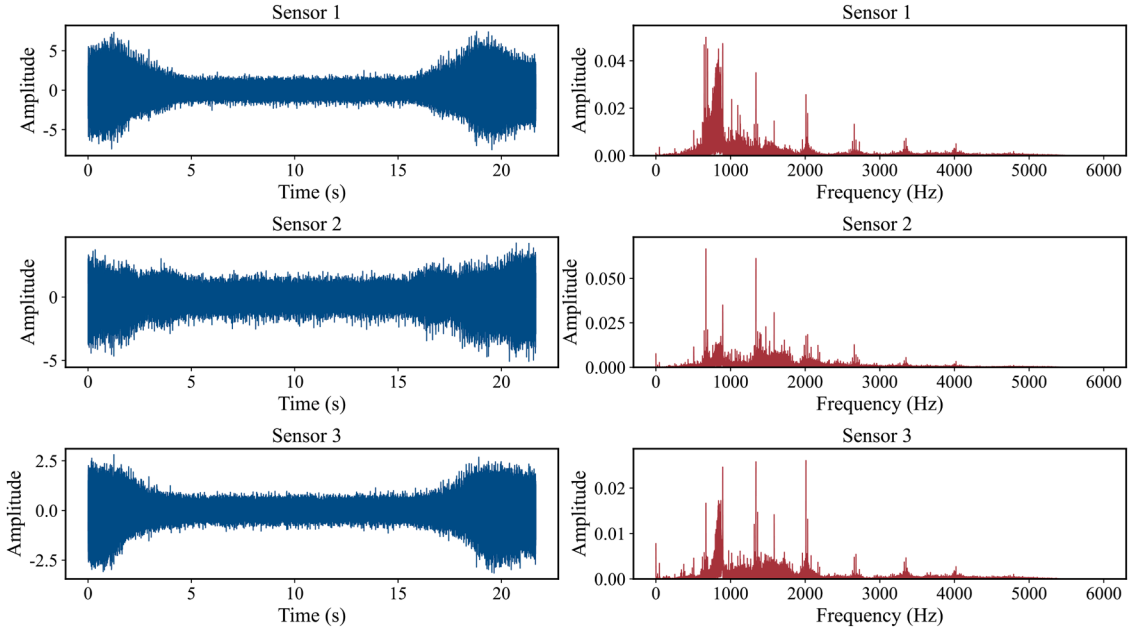
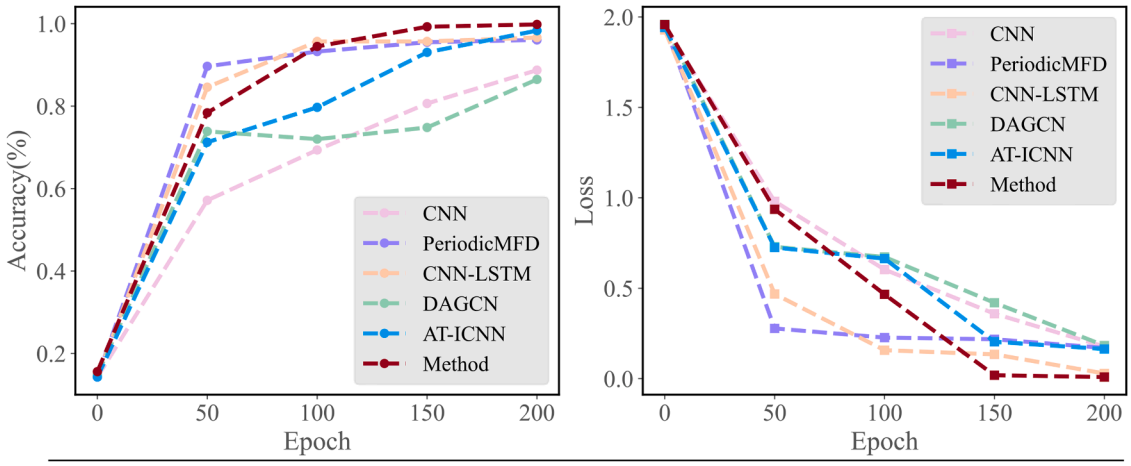


Fig. 12. Visualization of THU vibration signal and its frequency-domain curve.



Method	CNN	PeriodicMFD	CNN-LSTM	DAGCN	AT-ICNN	Method
Accuracy	88.43±0.01	96.03±0.002	97.01±0.03	84.34±0.02	97.67±0.06	**99.78±0.001**

Fig. 13. Comparison of training accuracy and loss among different models.

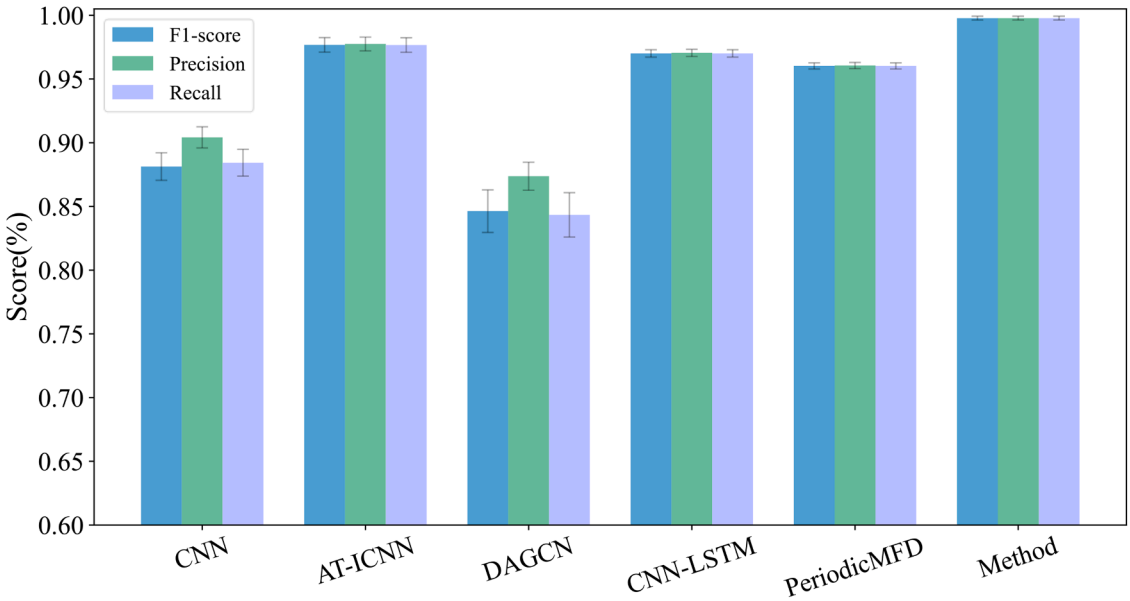


Fig. 14. Evaluation metrics of different comparative methods.

operating condition dataset [40], which contains combined bearing and gear faults, is used for validation. The specific fault combinations are listed in Table 3. Table 4 provides detailed parameters of the test bearings. All defects were introduced by laser etching with a machining accuracy of 0.01 mm. The experimental rig's detailed structure is shown in Fig. 11. To accurately measure vibration signals under time-varying speeds, the test rig is equipped with two triaxial accelerometers (TES001V). Under a load of 10 Nm and a sampling frequency of 12.8 kHz, the motor speed gradually rises from 0 to 2000 r/min during operation.

Compared to the NEFU dataset, the THU dataset introduces additional load under more complex time-varying speed conditions. Similarly, the bearing vibration signal with label 2 is selected for visualization, as shown in Fig. 12. In this case, 250 samples are used for each health status of the bearing.

A comparison with leading fault diagnosis approaches confirms the effectiveness of the proposed method, with its average accuracy and standard deviation presented in Fig. 13. With its powerful dynamic receptive field, the proposed method efficiently captures fault features under time-varying speeds.

Relative to the best and worst performing models, the proposed approach achieves improvements of 15.44 % and 2.11 %, respectively, demonstrating its strong diagnostic effectiveness. Experimental details are provided in Fig. 14. Fig. 15 illustrates the

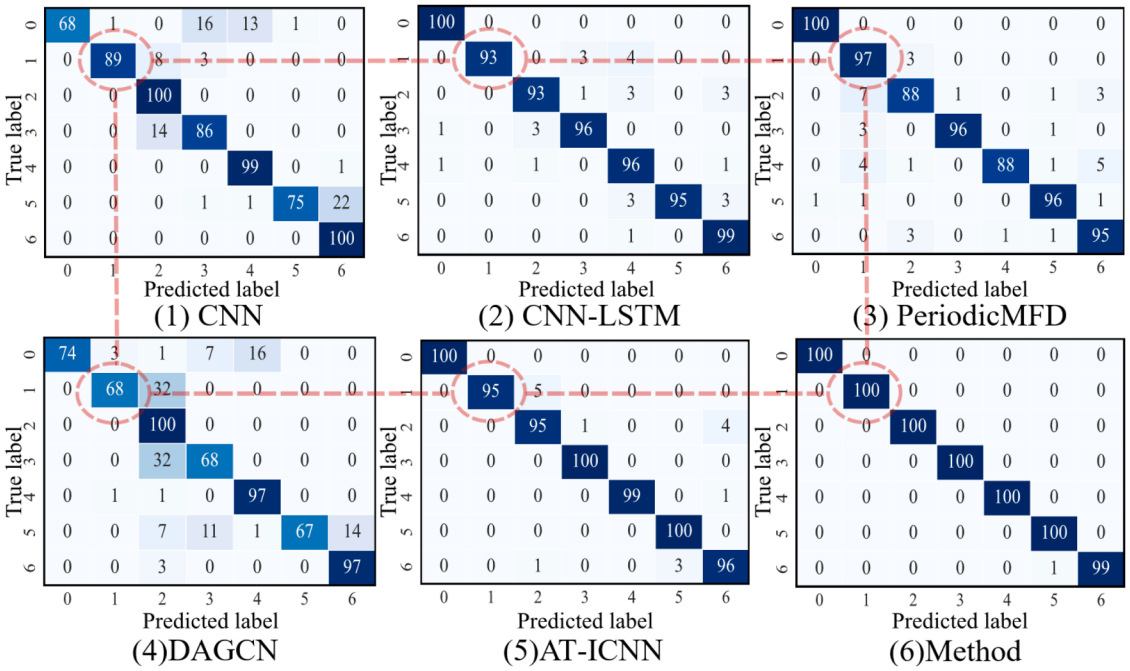


Fig. 15. Models confusion matrix analysis.

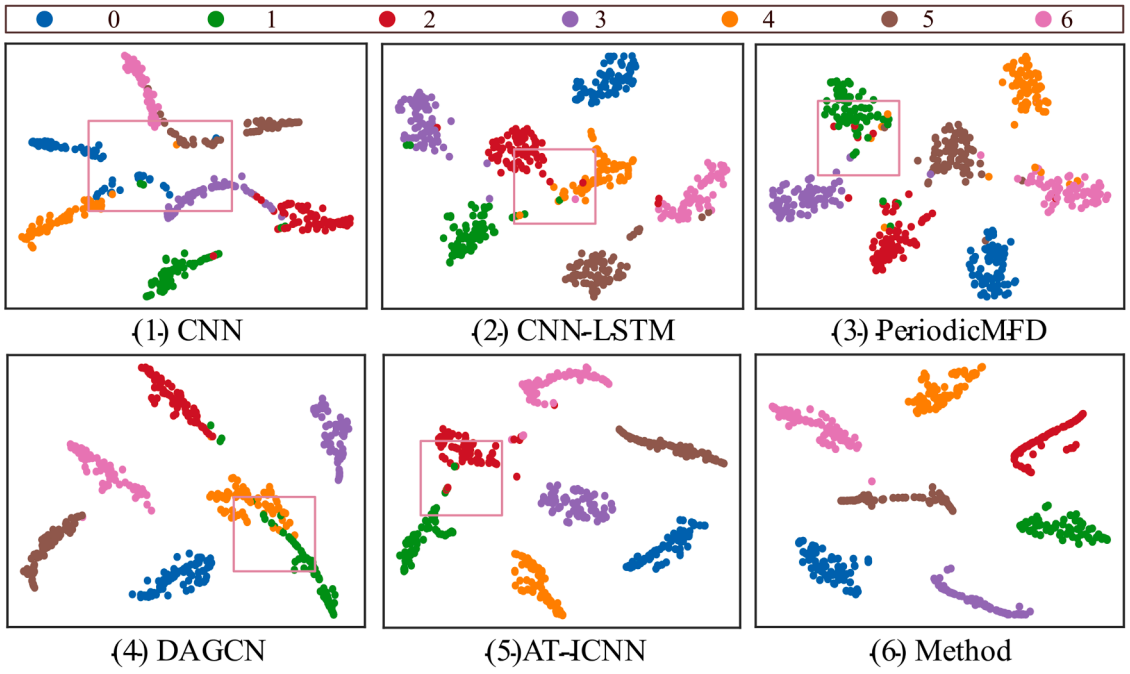


Fig. 16. t-SNE visualization of different models.

confusion matrices of the proposed method under different variable-speed conditions. It can be clearly observed that the proposed model achieves high classification accuracy for each type of fault sample. The low values of the off-diagonal elements indicate that misclassification occurs rarely. As shown in Fig. 16, the t-SNE visualization results under different comparison methods all exhibit certain intra-class aggregation. In contrast, the proposed method forms a more compact clustering space for samples of the same class, while the separation between different classes is more pronounced.

The ablation experiment results on the THU dataset are shown in Table 5. The Baseline-CNN achieved an accuracy of 88.43 %,

Table 5
Ablation experiments on THU.

Method	Accuracy(%)	F1-score(%)	Recall(%)
Baseline-CNN	88.43	88.78	88.92
ESC-only	92.10	92.05	92.33
DGCF-only	94.61	93.68	94.27
ESC-DGCN w/o Multi-Scale GCN	97.26	97.22	97.19
ESC-DGCN	**99.78**	**99.70**	**99.62**

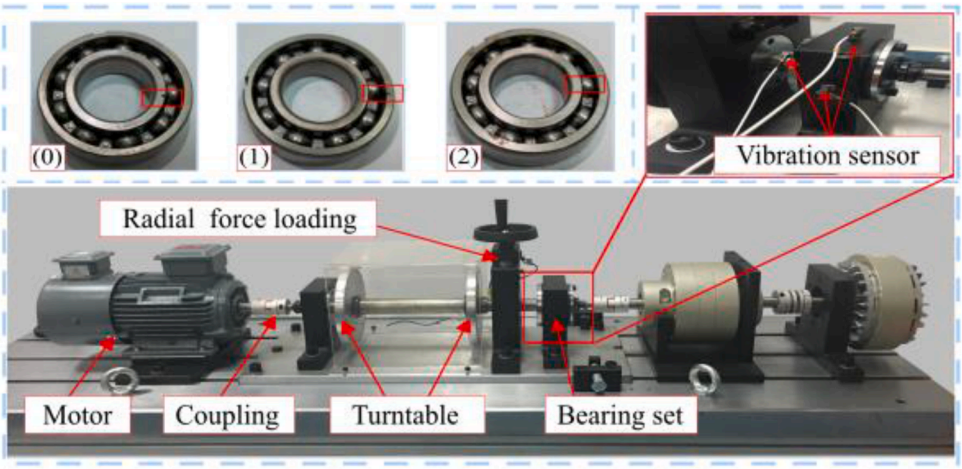


Fig. 17. Detailed structure of the SDUST fault diagnosis test rig [41].

Table 6
Basic parameters of the SDUST bearing.

Bearing Model	Inner Diameter	Outer Diameter	Contact Angle	Ball Diameter
6205	25 (mm)	52(mm)	0	7.938

Table 7
SDUST bearing dataset overview.

Health Condition	Label	Health Condition	Label
Healthy Bearing	0	Outer Race Fault	2
Inner Race Fault	1	Ball Fault	3

indicating limited performance. With the introduction of the ESC module, the accuracy increased to 92.10 %, suggesting that elastic convolution still provides stable gains on this dataset. The DGCF-only module reached 94.61 %, further validating the role of dynamic graph structure learning in modeling complex signals. The ESC-DGCN w/o Multi-Scale GCN experiment achieved 97.26 %, highlighting the strong advantage of combining the two core modules. Finally, the proposed ESC-DGCN demonstrated outstanding performance, fully indicating that the method possesses robustness and superior generalization capability under different data conditions.

4.4. Shandong university of science and technology time-varying speed bearing dataset

To further validate the model's generalization ability, we selected the Shandong University of Science and Technology (SDUST) time-varying speed bearing dataset [41], with a load of 40 Nm and a motor speed varying randomly from 1000 r/min to 2000 r/min. The experimental setup is composed of an AC motor, a speed controller, test bearings, a gearbox, a load control handle, and a magnetic powder brake. The configuration of the test rig is depicted in Fig. 17. The experimental bearing model is 6205, with a sampling frequency set to 25.6 kHz. The specific parameters and health statuses corresponding to the labels are provided in Tables 6 and 7.

Compared to the THU dataset, the SDUST dataset imposes more challenging conditions by introducing random speed variation under additional load. Similarly, the bearing vibration signal with label 2 is selected for visualization, as shown in Fig. 18. A total of 250 samples are used for each bearing condition.

The performance of ESC-DGCN was further validated on the SDUST dataset and compared with five widely studied fault diagnosis

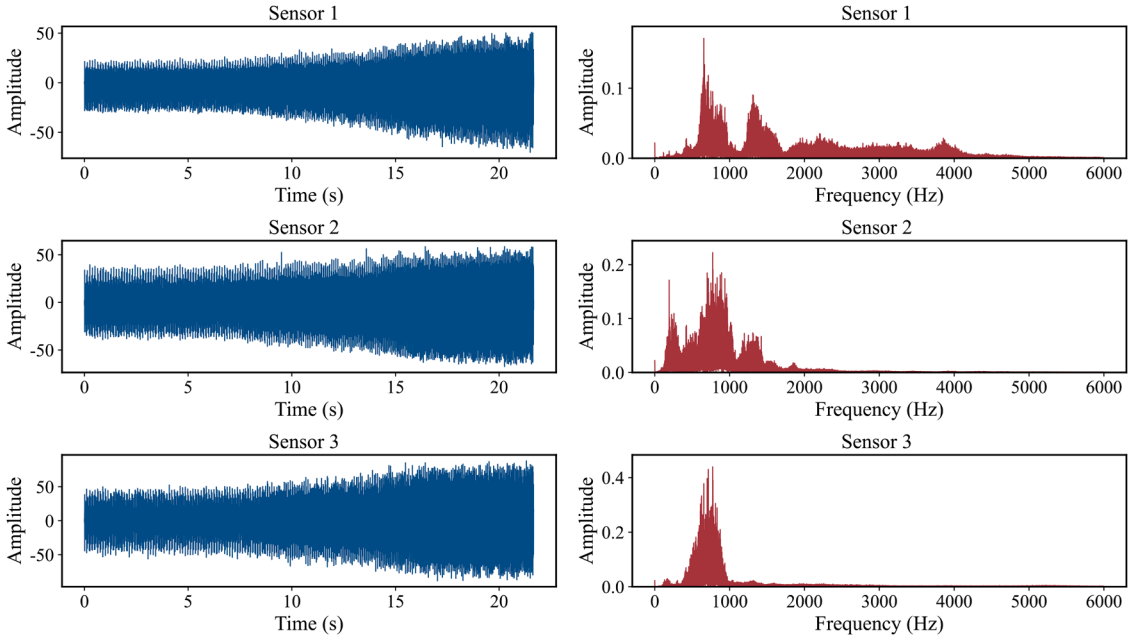
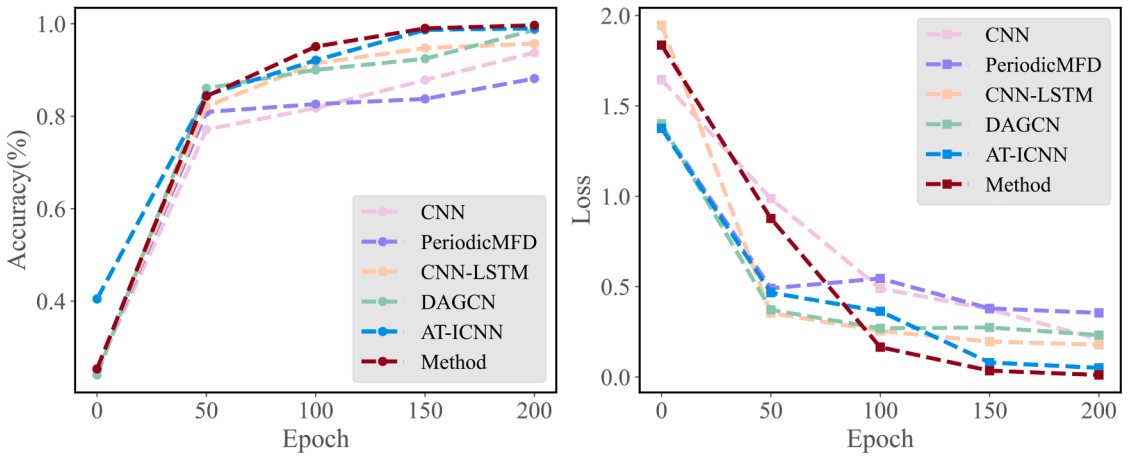


Fig. 18. Visualization of SDUST vibration signal and its frequency-domain curve.



Method	CNN	PeriodicMFD	CNN-LSTM	DAGCN	AT-ICNN	Method
Accuracy	92.09±0.04	89.79±0.02	94.75±0.01	97.92±0.01	98.33±0.04	**99.63±0.002**

Fig. 19. Comparison of training accuracy and loss among different models.

models. As shown in Fig. 19, the diagnostic accuracies of CNN, PeriodicMFD, CNN-LSTM, DAGCN, AT-ICNN, and the proposed ESC-DGCN are 92.09 %, 89.79 %, 94.75 %, 97.92 %, 98.33 %, and 99.63 %, respectively. Additional evaluation metrics, including F1-score, precision, and recall, are presented in Fig. 20.

Finally, the confusion matrix was employed for visualization, and the results are shown in Fig. 21. For fault type labeled as 3, the classification accuracy of different comparison methods remains below 98 %, indicating that the features of this fault are more difficult to distinguish under complex operating conditions. Furthermore, as illustrated in the t-SNE visualization results in Fig. 22, although all methods exhibit a certain degree of intra-class aggregation, the clustering boundaries for label 3 still show some overlap. In contrast, the proposed method achieves a more compact clustering distribution for samples of the same class, while the separation between different classes is more pronounced. The experimental results demonstrate that the proposed ESC-DGCN achieves high classification accuracy even under high-load and randomly varying speed conditions, indicating its strong capability in handling various types of bearing faults and compound fault diagnosis tasks.

The ablation experiment results on the SDUST dataset are presented in Table 8. The Baseline-CNN achieved an accuracy of 92.09 %,

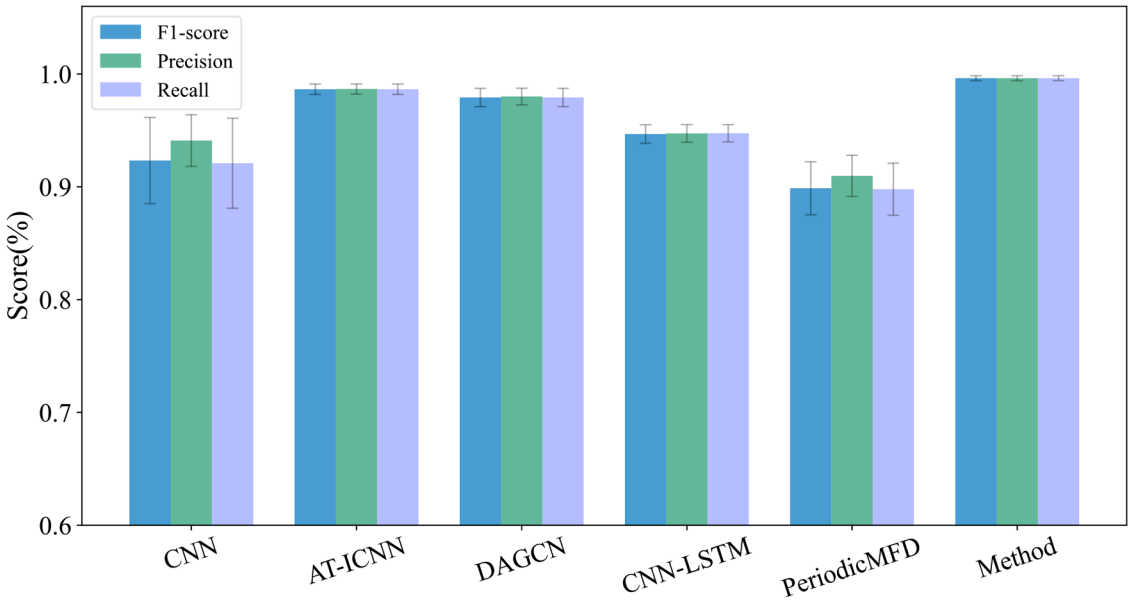


Fig. 20. Evaluation metrics of different comparative methods.

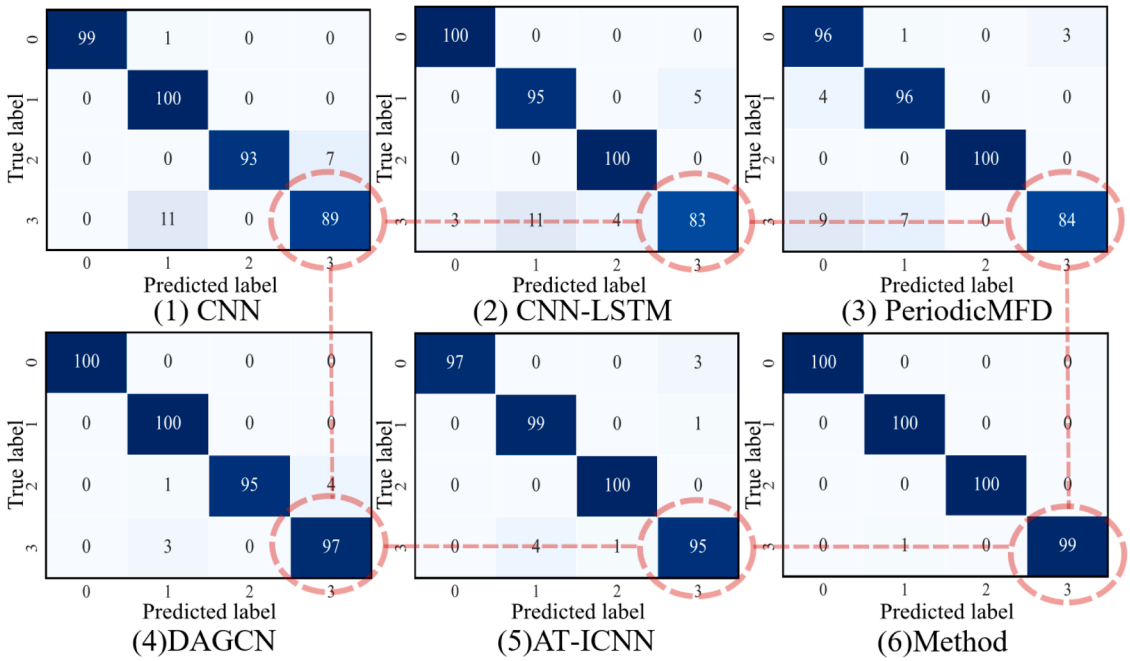


Fig. 21. Models confusion matrix analysis.

which is higher than the baseline levels on the previous two datasets but still shows noticeable performance limitations. The ESC-only model improved the accuracy to 95.24 %, indicating that elastic convolution can effectively enhance feature representation across different data environments. The DGCF-only model achieved 96.38 %, further confirming the advantages of dynamic graph modeling. When the two were combined (ESC-DGCN w/o Multi-Scale GCN), the accuracy increased to 98.11 %, showing a clear advantage over each individual module. Finally, the complete ESC-DGCN achieved an accuracy of 99.63 % on the SDUST dataset, with nearly perfect F1-score and Recall, once again validating the effectiveness and robustness of combining ESC with DGCN.

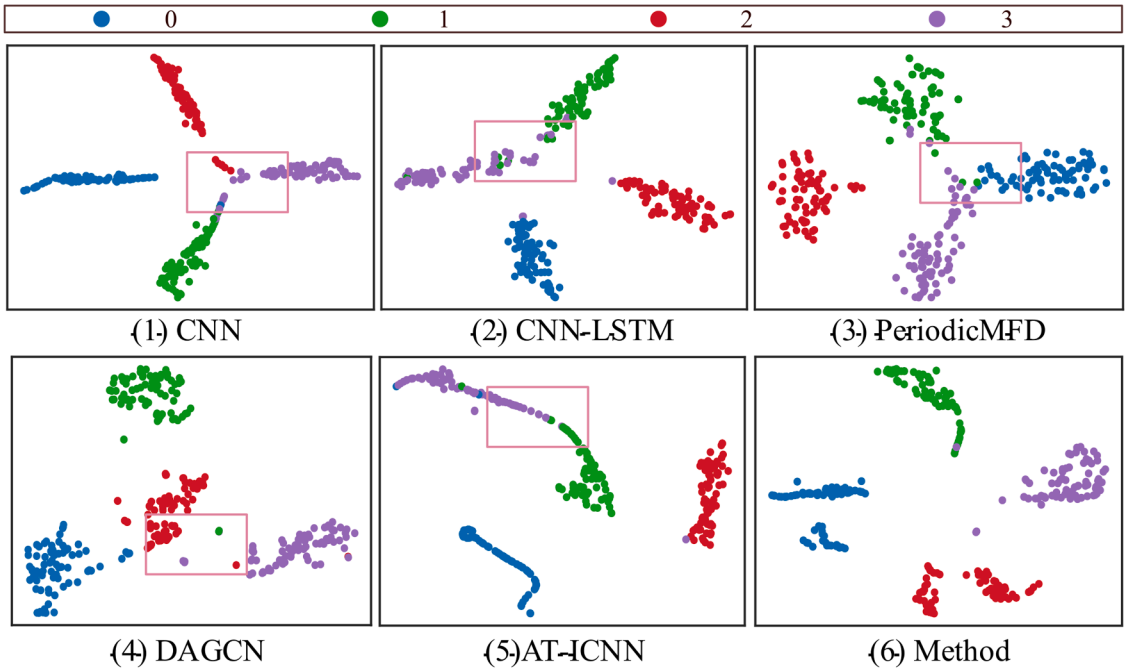


Fig. 22. t-SNE visualization of different models.

Table 8
Ablation experiments on SDUST.

Method	Accuracy(%)	F1-score(%)	Recall(%)
Baseline-CNN	92.09	91.67	92.17
ESC-only	95.24	94.85	95.01
DGCF-only	96.38	96.02	96.12
ESC-DGCN w/o Multi-Scale GCN	98.11	97.83	98.36
ESC-DGCN	**99.63**	**99.14**	**99.37**

5. Conclusion

To overcome the limitations of existing studies and the challenges posed by time-varying speed conditions in practical industrial environments, we propose a novel intelligent fault diagnosis framework, ESC-DGCN. The proposed method integrates convolution kernels with dynamic receptive fields and a strategy for adaptively constructing graph structures, which effectively reduces the dependence on single-sensor data and significantly improves fault diagnosis performance under non-Stationary conditions. In addition, the framework fuses input information from multiple sensors, thereby enhancing the model's adaptability to complex operating conditions. We evaluated the model's diagnostic performance using three different datasets: the NEFU bearing dataset under 0 Nm load and variable speed, the THU public bearing dataset under 10 Nm load and variable speed, and the SDUST public bearing dataset under 40 Nm load and randomly varying speed. Experimental results demonstrate the superiority of the proposed method over other state-of-the-art approaches.

Future research will focus on addressing challenges such as abrupt condition changes and class imbalance in real-world industrial applications. Meanwhile, we plan to further explore lightweight model structures and extend the proposed framework to a wider range of rotating machinery as well as cross-equipment, multi-source heterogeneous data scenarios to validate its generalization capability and practical engineering value.

CRediT authorship contribution statement

Xueyi Li: Visualization, Validation, Formal analysis, Data curation, Conceptualization. **Guangsheng Ran:** Writing – review & editing, Writing – original draft, Visualization, Validation. **Shuquan Xiao:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology. **Kaiyu Su:** Methodology, Investigation, Conceptualization. **Qi Li:** Supervision, Software, Resources, Conceptualization. **Tianyang Wang:** Project administration, Methodology, Funding acquisition, Conceptualization. **Zhaoye Qin:** Validation, Supervision, Software, Resources, Investigation, Funding acquisition. **Fulei Chu:** Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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